CS247 Assignment 1 — Slice Viewer

How to run

```
cd Assignment_1
cmake -S . -B build
cmake --build build -j
cd build && ./assignment
```

Key	Action
1 / 2	load lobster / head dataset
A	switch axis (X ⟶ Y ⟶ Z)
W / S	next / previous slice
B	background colour
Esc	quit

What it does

Loads a 16-bit CT volume into one `GL_TEXTURE_3D` and draws a full-screen quad. The fragment shader samples the 3D texture, so I never re-upload data when the slice or axis changes — I only change two `mat4` uniforms per frame.

Datasets: lobster 120×120×34, skewed_head 184×256×170.

Implementation

Geometry. One quad in NDC, built once in `setup()` (VAO + VBO + EBO, 6 floats per vertex = position + tex coord). Nothing about it changes at runtime.

Picking the slice (`buildTexMatrix`). Each vertex carries `(u,v,0)`. A `texTransform` matrix maps it to the 3D sample location, with the fixed axis set to the current slice:

axis	plane	maps (u,v,0) ⟶
0	Y–Z	(sliceX, u, v)
1	X–Z	(u, sliceY, v)
2	X–Y	(u, v, sliceZ)

I use `(slice+0.5)/dim` so it samples the voxel centre, which matches `GL_LINEAR` filtering.

Aspect ratio (`buildModelMatrix`). Compare the slice aspect to the window aspect and shrink the wider side:

```
if (winAspect > sliceAspect) sx = sliceAspect / winAspect;
else                         sy = winAspect   / sliceAspect;
```

`framebufferSizeCallback` updates the viewport on resize (uses framebuffer size, so retina is fine).

Shaders. Vertex shader applies `model` and `texTransform`; fragment shader does `texture(volume, texCoord).r` and writes it as greyscale.

Results

See `demo.mov` — loading both datasets, axis switching, slice scrubbing, and resizing without distortion.

Built and tested on Apple Silicon, OpenGL 4.1 core profile.